

CRF State Summary

Complete Framework State for Cross-Session Use

All results through Paper v4 + EIC Bridge + Track A.

Use this document to restore full context in any session.

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1. The Single Primitive

δ (Distinction) is CRF's sole undefined term. Declared, not proved. All other quantities derive from δ .

2. The Derivation Chain

$$\delta \rightarrow P \rightarrow \mathcal{C} \equiv \Pr(s|\mathcal{C}) \rightarrow D_{\text{KL}} \rightarrow \theta = 1 - e^{-D_{\text{KL}}/D_0} \rightarrow \mathcal{R} \rightarrow t = |\mathcal{R}_{\text{events}}|$$

$$\alpha = \ln(2), \quad \kappa = \frac{k_B \ln(2)}{H_{\text{mat}}}, \quad \tau = \tau_0 e^{-\kappa N_{\text{eff}}}, \quad N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$$

$$\Pr(s|\mathcal{C})|_{\mathcal{C}_{\min}} = |\langle s|\psi\rangle|^2 \quad [\text{Born Rule: Theorem via Gleason} + \text{SO}(3)]$$

$$\mathcal{C}(r) = \frac{\alpha}{1 - r_s/r} \Rightarrow ds_{\text{Schwarzschild}}^2 \quad G = \frac{\ln(2) \cdot c^2}{D_0} \quad T_{\mu\nu} = \mathcal{K}(\partial_\mu \mathcal{C} \otimes \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2)$$

$$G(t) \propto \Lambda(t) \propto \frac{1}{P(t)} \quad [\text{Cosmic co-variation, DESI 2024 consistent}]$$

3. Grand Equation v9.0

$$E_{v9.0} = \underbrace{A(F(\nabla \Pr(D(R(P))))))}_{\text{Samsara}} \oplus \underbrace{\delta_{\text{ind}}(W_{\text{max}}) \cdot \mathcal{A}_{\text{mass}}}_{\text{Barami}} \otimes \underbrace{\text{Resonance}(\mu, D_{\text{KL}})}_{\text{Dhamma}} \otimes \underbrace{\text{Sync}(D_{\text{KL},ij})^T}_{\text{The Loop}}$$

At $T = 1$: $E_{v9.0} = \delta$. Cascade closed.

4. EIC Bridge

CRF $\equiv$ EIC: Central Identification

$$\Omega(x, t) \equiv D_{\text{KL}}(\mathcal{C}(x, t) \| P)$$

EIC field equation derives entirely from δ -chain. All 6 terms have CRF origin. No free parameters remain.

EIC SINDy confirms: $I_{\text{eff}} = 0.7007 \approx \ln(2) = 0.6931$ (gap 0.76% = ε_0). EIC critical threshold $\alpha_c = 0.525 \equiv \theta$.

5. Complete Status Table

Item	Status	Source
δ as Indefinable Primitive	Declared	Session 10
$\Pr(s)$ from δ -frequency	Derived	Sessions 1–3
$\mathcal{C} \equiv \Pr(s \mathcal{C})$	Derived	Session 4
$\theta = 1 - e^{-D_{\text{KL}}/D_0}$	Derived	Session 10
3D space from $\text{SO}(3)$	Derived	Session 5
$\alpha = \ln(2)$	Derived	Session 8
$\tau = \tau_0 e^{-\kappa N_{\text{eff}}} \text{ (P0)}$	Derived	Sessions 6–8
Lemmas 13–15	Derived	Session 9
Born Rule via Gleason	Theorem	Session 11 / Paper v4
Schwarzschild metric	Derived	Paper v4
$G = \ln(2)c^2/D_0$	Derived	Paper v4
$T_{\mu\nu}$ from $\mathcal{C}$ -gradients	Derived	Paper v4
$G(t) \propto \Lambda(t) \propto 1/P(t)$	Prediction	Paper v4
$\Omega \equiv D_{\text{KL}}$ (EIC bridge)	Derived	CRF-EIC Bridge
$I_{\text{eff}} \approx \ln(2)$ (EIC)	Confirmed	Track B.1
$\alpha_c = 0.525 \equiv \theta$	Derived	CRF-EIC Bridge
$\gamma \propto \ln(2)$	Derived	Track A
$\gamma = \ln(2)/\theta_{\text{crit}}$	Hypothesis	Track A
$\Delta I \cdot \Delta \Pi \geq \ln(2)/2$	Derived	Track A
$\hbar \propto \ln(2) \cdot D_0$	Hypothesis	Track A
$\hbar$ SI bridge	Open	Needs P0 + D_0
$d_s \rightarrow 3$	Open	Needs $N \gg 10^6$
$\text{SO}(3)$ minimality formal proof	Open	#16
Kerr metric	Open	Broken $\text{SO}(3)$
P0 experiment	Ready	SNSPD protocol

6. Maturity: Level 6/8

Dimension	Score
Internal Consistency	5.5/8
Mathematical Rigor	5.5/8
Predictive Power	6.0/8
Empirical Grounding	2.0/8
Philosophical Foundation	6.5/8
Cross-domain Coverage	6.5/8
Overall	6.0/8

Gate to Level 7: P0 experiment ($\tau_1 < \tau_2$ at $> 5\sigma$).

Gate to Level 8: $G(t)$ variation via gravitational wave standard sirens.

7. Honest Boundaries

What CRF Does Not Claim

- Does not solve the hard problem of consciousness
- Does not claim $\hbar_{\text{CRF}} = \hbar_{\text{SI}}$ numerically — only structurally
- Does not claim $d_s = 3$ proved — only that $\text{SO}(3)$ selects it
- $G(t) \propto \Lambda(t)$ is a prediction, not confirmed
- γ and $\hbar$ are hypothesis, not derived

8. Key Vocabulary

Term	Meaning
Derived	From axioms, chain forced, no alternative
Hypothesis	Consistent, correct structure, needs formal bridge
Confirmed	Empirical evidence supports
Open	Gap, needs derivation or experiment
Watching	Interesting, not yet CRF claim

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